FermatPlanes: Easy-Arrow Linear Planes, Four-Factor Algebraic Identities, and Coefficient Span on Fermat Fourfolds in Lean 4

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ORCID: 0009-0006-7068-3718 | GitHub: BenFrohman/FermatPlanes
Dedicated 1-to-1 Zenodo DOI: 10.5281/zenodo.23652978

Earliest Commit Provenance: 2026-09-21 04:30:49 AM CDT (UTC-5)
Siloed Archival Edition: October 25, 2026 (America/Chicago CDT)

Abstract

Standalone Lean 4 formalization (‘FermatPlanes/EasyArrow.lean‘, ‘FermatDatum.lean‘, ‘FermatIdentities.lean‘, ‘FourFactorIdentity.lean‘, ‘LinearPlanes.lean‘, ‘Membership.lean‘, ‘SexticPair.lean‘) of explicit algebraic 2-planes contained in Fermat fourfolds and the four-factor polynomial factorization identities governing their cycle classes.

1 Scope, Scientific Honesty, and Non-Claim Boundary

Proves the ’easy arrow’ (explicit linear planes define algebraic cycles and therefore rational Hodge classes) and associated polynomial identities in Lean 4. Explicitly does not claim the converse Hodge Conjecture.

- **Repository:** `https://github.com/BenFrohman/FermatPlanes`
- **Dedicated 1-to-1 Zenodo DOI:** `10.5281/zenodo.23652978`
- **Earliest Git Commit SHA:** `0efda24706acd35a568d2e95141ac406b2a3de1d` (2026-09-21 04:30:49 AM CDT (UTC-5))
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2 Technical Overview & Repository Specification

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Easy-arrow planes on the Fermat quartic fourfold
X : z ^4 + + z ^4 = 0 ^5
with ^4 = -1. Remainder identity
z^4 + w^4 = (z − w)(cubic) + (1+ ^4) w^4
vanishes on the locus ^4 = -1. This is **not** the Hodge conjecture
and **not** a global constructor.
- [HODGE](https://github.com/BenFrohman/HODGE) — fourfold Hodge sentence; Cycle-Section on named hosts; ‘general_fourfold‘ open

- [NoetherLefschetz](https://github.com/BenFrohman/NoetherLefschetz) — very general high-degree hosts; no extra classes

- [CubicFourfold](https://github.com/BenFrohman/CubicFourfold) — cubic Hodge numbers; Zucker cited

- [StandardConjectures](https://github.com/BenFrohman/StandardConjectures) — A/B/C/D skeleton

- [TateConjecture](https://github.com/BenFrohman/TateConjecture) — Tate sentence only

- [DerivedCategories](https://github.com/BenFrohman/DerivedCategories) — D^b notes

- [HODGE-DISPROOF](https://github.com/BenFrohman/HODGE-DISPROOF) — disproof *type*, not a disproof

Copies remain under ‘Hodge/Attempt/‘ in HODGE. They are not imported by ‘Hodge.lean‘.

3 Cryptographic Provenance & File Inventory

Every source artifact in `BenFrohman/FermatPlanes` is audited via `titan-audit` (`audit/TITAN_AUDIT_CERTIFI
and sealed under a deterministic SHA-256 Merkle tree via `sealguard` (`PROVENANCE_MERKLE_ROOT.json`) signed with Ed25519 key fingerprint `b428fb72676f1c52`.

Repository Path	Size	SHA-256 Prefix
COEFFICIENTS_142.md	1456 B	a9ada0eb44ba4f2d...
COPYRIGHT	347 B	68c81d3167dcc3aa...
EQUATIONS.md	1944 B	99c9439fc55fb6dd...
EasyArrow.lean	1000 B	7457ca90e22f9552...
FERMAT_PLANES.md	939 B	a8007abb1c35c583...
FermatDatum.lean	886 B	214a13ab3ec5904e...
FermatIdentities.lean	2560 B	1f91cff714d9ce14...
FourFactorIdentity.lean	1101 B	2f05b0184eea843b...
LICENSE	11355 B	9da72c2bee4f20ba...
LINEAR_SPAN.md	1465 B	d23a94590fc22c48...
LinearPlanes.lean	1376 B	4b10067cd14f5f71...
Membership.lean	1147 B	2aff6f9c0cb48b21...
NOTICE	324 B	36fb4e0c499e58cc...
README.md	1295 B	d862427a8342b3d1...
SexticPair.lean	1944 B	1ec7bf15041b5b95...
docs/CONCRETE_S_REJECT.md	959 B	ec0e5caba3a7955d...

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